Use-Case Flow Specification

Urban Rooftop Gardening Co-op App, Problem Statement #65

Use Case: UC-04 **List Surplus Produce for Trade (FR-004)**

Primary Actor: Co-op Gardener

Preconditions

- Gardener has an account and is logged in.
- Gardener owns at least one approved rooftop plot (UC-01).
- The crop being listed was already logged through UC-02, so the system knows how much was harvested.

Postconditions

- The listing shows up on the shared trade board for other members to see.
- Trade board reflects the new listing right away, per NFR-001.

Main Success Scenario

1. Gardener taps “List Surplus Produce” from the home screen. The app opens a short form asking for crop type, quantity, and how long the listing should stay up.

2. Gardener picks the crop from their own logged crops (so they can't accidentally list something they never grew) and types in how much they want to trade.

3. Gardener picks a start and end date for the listing.

4. App checks two things in the background: that the quantity isn't more than what was actually harvested, and that the dates make sense (end date isn't before start date, nothing in the past).

5. Gardener reviews the details and hits submit.

6. The listing is saved, and this automatically triggers «include» Update Trade Board, so every other member's board updates without them needing to refresh anything.

7. Gardener sees a confirmation screen with the listing summary.

8. Done, use case ends successfully.

Alternate Flow

4a. Quantity or Dates Don't Check Out

- 4a1. App notices gardener entered more produce than they actually logged, or the dates are backwards/ expired.
- 4a2. A short error message pops up next to the field that's wrong, so the gardener knows exactly what to fix.
- 4a3. Gardener edits the value and tries submitting again.
- 4a4. Once it checks out, the flow just picks back up at step 5. If the gardener decides to bail instead, nothing gets listed and the use case simply ends.